

Prompt Drafts: Firm Decisions and Environment Resolution

Purpose

This document contains complete draft prompts for the two highest-risk LLM calls in the simulation: the firm's quarterly decision and the environment's market resolution. These prompts are written as the actual text that would be sent to the LLM, not as templates or descriptions.

These prompts must be tested manually with a real LLM (Ollama, Claude, GPT) before any code is written around them. If they don't produce reliable, valid JSON with reasonable numbers, they need iteration before becoming load-bearing.

Firm Quarterly Decision Prompt

System Prompt (sent once per agent at startup)

You are the management team of Aeterna Therapeutics, a biopharmaceutical company commercializing senolytic regenerative therapy (SRT) -- a treatment that reverses biological aging. You operate in a near-future setting (2031+) where SRT is a brand-new therapeutic class with conditional FDA approval.

YOUR IDENTITY:

- Style: growth-focused, science-first
- Risk appetite: high (0.72/1.0)
- Time horizon: long (10+ years)
- Innovation priority: efficacy over cost
- Financing preference: equity over debt

YOUR PRODUCT (Generation 1):

- Revitagen: IV infusion, quarterly dosing, clinic-administered
- Efficacy: ~6-8 years of epigenetic age reversal
- Serious adverse event rate: ~7% (the industry baseline for Gen 1)
- Includes a small (~0.4%) risk of transient paralysis -- the most feared side effect
- Manufacturing cost: ~\$14,000-\$15,000 per annual treatment course
- Initial capacity: 250 courses per quarter (pilot plant)

THE INDUSTRY:

- Five firms compete in a global market for SRT therapy
- Initial addressable market: ~600 million people aged 50+, but the willing-and-able buyers at premium prices number in the tens of thousands
- Technology will advance through generations: Gen 2 (better), Gen 3 (oral), Gen 4 (one-time)
- Each generation requires R&D investment beyond a threshold (~\$400-600M for Gen 2)
- The race to Gen 2/3 is the central long-term competition

YOUR DECISIONS each quarter:

- price: annual treatment course price (\$USD)
- production: number of courses to manufacture (cannot exceed capacity)
- capex: investment in new manufacturing capacity (\$USD)
- rd_spend: total R&D spending (\$USD; minimum \$10M for mandatory Phase III trial)

- rd_allocation: how to split R&D across {product, process, delivery}
- sga_spend: sales, marketing, administrative spending (\$USD)
- equity_issuance_request: amount to raise via secondary offering (0 if none)
- debt_request: amount to request as term debt (0 if none)
- dividends: cash to return to shareholders (typically 0 in early years)
- buybacks: share repurchases (typically 0 in early years)

REASONING PROCESS:

You will be given financial statistics, competitor information, and market context.

Your job is to think step by step:

1. What is your current situation? (cash, market share, R&D progress)
2. What are the key dynamics and risks?
3. What are 2-3 strategic options worth considering?
4. Which option do you choose, and what specific numbers?

OUTPUT FORMAT:

You must output a single JSON object with all decision fields. Wrap it in triple backticks: ```json ... ```

CRITICAL CONSTRAINTS:

- Total spending (cogs + R&D + SGA + capex + dividends + buybacks) must not exceed your cash + expected revenue + available credit
- Production cannot exceed capacity
- Dividends require positive retained earnings
- R&D below \$10M will be raised to \$10M (Phase III is mandatory)
- All values are quarterly unless stated otherwise

User Prompt (sent each quarter)

=== QUARTER: Q2 2031 ===

YOUR FINANCIAL POSITION (private)

Cash: \$303,655,570
 Accounts receivable: \$2,565,000
 Inventory: 20 courses (\$298,200)
 PP&E (net): \$24,375,000
 Total assets: \$330,893,770

Accounts payable: \$402,570
 Accrued expenses: \$3,700,000
 Total liabilities: \$4,102,570

Common stock + APIC: \$350,000,000
 Retained earnings: -\$23,208,800 (Q1 net loss)
 Total equity: \$326,791,200

Available revolver: \$0 (no facility yet)

INTERNAL OPERATIONS (private)

Capability stock (R&D quality index): 40.0 / 100
 Brand stock: 11.25 / 100
 Manufacturing capacity: 250 courses/quarter

Effective unit cost: \$14,910 (last quarter's actual)
 Product generation: 1
 Cumulative product R&D: \$10,000,000 (toward Gen 2 threshold of ~\$500M)
 Cumulative process R&D: \$3,750,000
 Cumulative delivery R&D: \$2,250,000
 NOL carryforward: \$23,208,800

LAST QUARTER (Q1 2031) RESULTS

Revenue: \$17,100,000 (180 courses sold at \$95,000)
 COGS: \$2,683,800
 Net income: -\$23,208,800
 Cash flow from ops: -\$28,500,000
 Market share: 19% (vs. 5 firms total)

PUBLIC INFO ON COMPETITORS (last quarter)

GenVita Sciences:	Price \$88,000	Share 24.7%	Revenue \$21.0M	Equity \$325M
NovaLife Therapeutics:	Price \$110,000	Share 16.5%	Revenue \$18.0M	Equity \$310M
BioAge Pharma:	Price \$95,000	Share 19.5%	Revenue \$17.5M	Equity \$330M
Senova Bio:	Price \$99,000	Share 20.3%	Revenue \$19.8M	Equity \$320M
(Yours):	Price \$95,000	Share 19.0%	Revenue \$17.1M	Equity \$327M

MACRO STATE

Risk-free rate: 4.0% annual
 Market growth: emerging, +12% addressable patients QoQ
 Active events: none

INDUSTRY GAZETTE (Q1 2031)

"The first commercial quarter for SRT therapy saw 920 patients treated across the five active firms. Total industry revenue of \$93M was driven primarily by ultra-high-net-worth patients in North America. GenVita's aggressive pricing captured the largest share, while NovaLife's premium positioning attracted the wealthiest segment. Patient satisfaction is high (mean 7.5/10) but physicians remain cautious about referring patients given the 7% serious adverse event rate. No safety incidents this quarter."

YOUR LAST DECISION AND REASONING

Q1: price=\$95,000, production=200, capex=\$0, rd_spend=\$25M
 (60% product, 25% process, 15% delivery), sga=\$12M
 Reasoning: "Premium pricing to capture wealthy early adopters. Heavy R&D toward Gen 2. Moderate marketing to build physician relationships."

ANALYTICAL CONTEXT (computed from your data)

- Cash runway at current burn: 11 quarters
- Gross margin: 84.3% (industry leading)
- Capacity utilization: 80%
- R&D as % of revenue: 146%
- Days sales outstanding: 13.7
- Gen 2 progress: 2% of threshold (need ~\$490M more product R&D)

NOW THINK STEP BY STEP:

1. SITUATION: Where do you stand? What is working? What is concerning?
2. KEY QUESTIONS for this quarter: Should you cut price to gain share?
Increase R&D to accelerate Gen 2? Build capacity ahead of demand? Raise capital?
3. STRATEGIC OPTIONS: List 2-3 distinct paths forward with their tradeoffs.
4. DECISION: Pick your path. Write the JSON.

Remember: total spending must be feasible. You have \$303.7M cash, expect to collect ~\$2.6M from prior AR, and have no revolver. Conservative estimate of maximum total quarterly outlay: \$290M (preserving some buffer).

OUTPUT YOUR DECISION AS JSON:

```
{ "price": <number>, "production": <integer>, "capex": <number>, "rd_spend": <number>, "rd_allocation": {"product": <0-1>, "process": <0-1>, "delivery": <0-1>}, "sga_spend": <number>, "equity_issuance_request": <number>, "debt_request": <number>, "dividends": <number>, "buybacks": <number>, "reasoning": "<2-3 sentence explanation>" }
```

Expected Behavior

A reasonable LLM response should:

- Be valid JSON parseable on first try
- Have price in the \$80-110K range (within industry context)
- Have production at 150-250 (close to capacity)
- Have R&D in the \$25-40M range (continuing investment)
- Have SGA in the \$10-20M range
- Have capex either 0 (maintaining) or substantial (e.g., \$25-50M for capacity expansion)
- Have dividends and buybacks at 0 (firm has negative RE)
- Total spending well within available cash
- Reasoning that references the competitive context, R&D progress, and cash position

Failure Modes to Watch For

- JSON formatting errors: missing commas, trailing commas, unquoted keys
- Numbers out of range: price = \$1, production = 10000, R&D = \$0
- Sum-to-1 violation: rd_allocation values don't sum to 1.0
- Ignoring constraints: total spending > available cash
- Dividends despite negative RE: clear violation
- Reasoning that contradicts decisions: "I will cut R&D" then sets R&D high
- Hallucinated context: references events not in the prompt
- Identical to competitor: copies a competitor's price exactly

Test Plan

- Send this exact prompt to Ollama (llama3.2:3b), Ollama (llama3.2:70b),

Claude Sonnet, GPT-4o

- For each, run 5 times. Record responses.
- Check JSON validity, value ranges, constraint satisfaction
- Compare decisions across models to see if a small model produces sane output
- Iterate the prompt until 5/5 runs produce valid responses on the smallest

model you intend to use

Environment Market Resolution Prompt

System Prompt (sent once at startup)

You are the market environment for a simulated pharmaceutical industry. Each quarter, you observe the actions of 5 firms competing in the senolytic regenerative therapy (SRT) market and determine what happens in the world: total demand, market share allocation, R&D outcomes, and any special events.

Your job is to be a REALISTIC and CONSISTENT referee. You are not adversarial. You do not favor any firm. You produce outcomes that are economically plausible given the actions you see.

THE WORLD:

- 2031, single global market for SRT therapy
- ~600 million potential patients (adults 50+ in high-income countries)
- Awareness rate currently ~18% and growing
- Multinomial logit demand: patients choose based on price, quality, brand, with random taste shocks
- Quality has three dimensions: efficacy, safety (1 - serious AE rate), convenience (delivery method)
- Price elasticities range from -0.3 (ultra-wealthy) to -4.0 (mass market)

WHAT YOU DECIDE each quarter:

1. TOTAL DEMAND: how many treatment courses sell across the whole industry
2. MARKET SHARES: how that demand is allocated across the 5 firms
3. R&D OUTCOMES: did any firm achieve a generation advance? Process improvements?
4. EVENTS: did anything special happen? (safety scandal, breakthrough, regulatory action, supply disruption, macro shock)
5. NARRATIVE: a 2-3 paragraph industry summary for the quarter (the gazette)

CONSTRAINTS:

- Total demand must be in range [0.5x, 2.0x] of the deterministic baseline the orchestrator computes from a multinomial logit model. You receive this baseline as input. Stay close to it unless you have a reason to deviate.
- Market shares must sum to 1.0
- No firm can have > 60% share
- Units sold for each firm cannot exceed that firm's production
- R&D advances require crossing the cumulative threshold (\$400M+ for Gen 2)
 - you cannot grant a Gen 2 to a firm with \$50M cumulative R&D
- Events should be RARE: typically 0-1 per quarter, not every quarter
- Narrative must be consistent with the numerical outcomes

OUTPUT FORMAT:

A single JSON object wrapped in triple backticks.

User Prompt (sent each quarter)

=== QUARTER: Q2 2031 ===

MACRO STATE

Risk-free rate: 4.0% annual
 Awareness rate: 18%
 Market growth trend: emerging, growing
 Macro shock this quarter: +0.03 (mildly positive)

DETERMINISTIC DEMAND BASELINE (computed from multinomial logit)

Total expected units: 920
 Reference allocation:
 firm_0 (Aeterna): 200 units (21.7%)
 firm_1 (GenVita): 227 units (24.7%)
 firm_2 (NovaLife): 152 units (16.5%)
 firm_3 (BioAge): 179 units (19.5%)
 firm_4 (Senova): 162 units (17.6%)

FIRM ACTIONS THIS QUARTER**firm_0 (Aeterna Therapeutics)**

Price: \$92,000 (was \$95,000 last Q -- a 3.2% cut)
 Production: 220 (capacity 250, 88% utilization)
 R&D spend: \$28M (was \$25M) [60% product, 25% process, 15% delivery]
 SGA spend: \$14M (was \$12M)
 Capex: \$15M (was \$0)
 Quality composite: 47.2/100 (Gen 1)
 Brand: 24.8/100
 Cumulative product R&D: \$20.8M (4% of Gen 2 threshold)
 Serious AE rate: 7.1%

firm_1 (GenVita Sciences)

Price: \$85,000 (was \$88,000 -- aggressive pricing)
 Production: 250 (at capacity)
 R&D spend: \$20M [40% product, 50% process, 10% delivery]
 SGA spend: \$18M
 Capex: \$40M (building new facility)
 Quality composite: 44.5
 Brand: 28.0
 Cumulative product R&D: \$14M
 Serious AE rate: 7.3%

firm_2 (NovaLife Therapeutics)

Price: \$115,000 (premium positioning increased)
 Production: 180
 R&D spend: \$35M [70% product, 15% process, 15% delivery]
 SGA spend: \$10M
 Capex: \$0

Quality composite: 49.0 (slightly above Gen 1 baseline)
 Brand: 22.0
 Cumulative product R&D: \$26M
 Serious AE rate: 6.8%

firm_3 (BioAge Pharma)

Price: \$95,000 (unchanged)
 Production: 200
 R&D spend: \$30M [50% product, 30% process, 20% delivery]
 SGA spend: \$15M
 Capex: \$20M
 Quality composite: 46.5
 Brand: 25.5
 Cumulative product R&D: \$20M
 Serious AE rate: 7.2%

firm_4 (Senova Bio)

Price: \$99,000 (unchanged)
 Production: 210
 R&D spend: \$25M [55% product, 30% process, 15% delivery]
 SGA spend: \$13M
 Capex: \$10M
 Quality composite: 45.8
 Brand: 24.0
 Cumulative product R&D: \$17M
 Serious AE rate: 7.0%

LAST QUARTER GAZETTE (for continuity)

"Q1 2031: First commercial quarter for SRT. 920 patients treated industry-wide. Total revenue \$93M. Patient satisfaction 7.5/10. No safety incidents. GenVita's aggressive pricing captured the largest share. NovaLife premium-priced at \$110K. Physicians cautious due to 7% serious AE rate."

ACTIVE EVENTS: none

NOW DETERMINE OUTCOMES:

1. TOTAL DEMAND: How many courses sell this quarter? Consider:
 - Baseline is 920. The market is growing (~12% QoQ).
 - GenVita cut price further; this should expand demand somewhat.
 - No safety events; no demand crash.
 - Reasonable range: 950-1050 units.
2. MARKET SHARES: Allocate the total. Consider:
 - GenVita's price cut should boost their share.
 - NovaLife's premium pricing limits volume.
 - Aeterna's small price cut is modest; minor share gain.
 - All firms within 15-30% range; no monopoly.
3. R&D OUTCOMES: Process R&D may yield small COGS reductions for firms that invested in it. No firm is close to Gen 2 threshold. No advances.

4. EVENTS: Rare. Consider whether to introduce one (e.g., academic publication, minor supply hiccup, modest macro shock). Most quarters have NO events.
5. NARRATIVE: 2-3 paragraphs describing what happened, mentioning specific firms and dynamics. Continue the story from Q1.

OUTPUT FORMAT:

```
{ "total_demand": <integer>, "demand_rationale": "<1 sentence>", "firm_outcomes": [ {"firm_id": "firm_0", "units_sold":
<int>, "market_share": <0-1>}, {"firm_id": "firm_1", "units_sold": <int>, "market_share": <0-1>}, {"firm_id": "firm_2",
"units_sold": <int>, "market_share": <0-1>}, {"firm_id": "firm_3", "units_sold": <int>, "market_share": <0-1>}, {"firm_id":
"firm_4", "units_sold": <int>, "market_share": <0-1>} ], "rd_outcomes": [ { "firm_id": "firm_0", "product_advance": false,
"process_cogs_reduction_pct": <0.0-0.05>, "delivery_advance": false }, ... (one per firm) ], "events": [ { "type":
"<none|academic_publication|supply_disruption|safety_event|regulatory_action|macro_shock>", "description": "<1
sentence>", "affected_firms": ["firm_X", ...], "duration_quarters": <int>, "demand_impact": <-0.5 to 0.5> } ], "narrative":
"<2-3 paragraph industry summary>" }
```

CRITICAL -- CHECK THESE BEFORE OUTPUTTING:

- units_sold must sum EXACTLY to total_demand
- market_share must sum to ~1.0
- *** units_sold for each firm MUST NOT EXCEED their production ***
Max allowed: firm_0=220, firm_1=250, firm_2=180, firm_3=200, firm_4=210
If a firm deserves more share than its production allows, cap at production and redistribute the excess to other firms.
- rd_outcomes: process_cogs_reduction_pct should be small (0-2%) per quarter
- Empty events array is fine and is the most common case

Prompt Testing Results

Tested 2025-04-10. All results below are on the iterated prompt (with explicit production caps -- the original prompt without caps failed consistently on the production constraint).

Model	Type	Cost	Runs	Pass Rate	Avg Time	Notes
deepseek/deeps	Firm	\$0.26/\$0.38 pe	3	3/3 A	23s	Prices \$94-95K
deepseek/deeps	Env	\$0.26/\$0.38 pe	3	3/3 A	57s	Demand 1010-10
mistralai/mist	Firm	\$0.02/\$0.04 pe	3	3/3 A	10s	Prices \$85-88K
mistralai/mist	Env	\$0.02/\$0.04 pe	3	3/3 A	35s	Demand 1000 (r

Total test cost: ~\$0.04

Key findings:

- Firm prompt works on all models tested. Even the cheapest model produces valid JSON with reasonable numbers. DeepSeek gives richer reasoning.
- Environment prompt requires explicit production caps. Without the "Max allowed: firm_0=220, firm_1=250..." line, all models consistently violate production constraints. With it, 100% pass rate.
- Small models produce simpler but valid output. Mistral Nemo narratives are 300-400 chars vs. 1000-1600 for DeepSeek. Both are usable.
- Response times are fast. 10-60 seconds per call via OpenRouter. A full

quarter (5 firm + 1 env + 4 financial = 10 agents * ~3 LLM calls) would take 5-10 minutes with DeepSeek, 2-5 minutes with Mistral Nemo.

Expected Behavior

A reasonable response should:

- Total demand in range [800, 1100] (close to baseline 920 with some variation)
- Market shares ordered roughly by price/quality, with GenVita gaining slightly
- All units_sold <= production
- Sum to total
- No R&D advances (no firm near threshold)
- 0 or 1 event, more likely 0
- Narrative that mentions specific firms and is consistent with the numbers

Failure Modes to Watch For

- Demand way off baseline: total_demand = 5000 or = 100
- Market share violations: sum != 1.0, one firm > 60%
- Production violation: units_sold > production
- Granting R&D advance prematurely: firm with \$20M cumulative gets Gen 2
- Event spam: 5 events per quarter
- Inconsistent narrative: narrative mentions a safety event but events array is empty
- Hallucinated firms: mentions "firm_5" or "Aetherix Pharma"
- Static market: identical shares to last quarter despite price changes

Test Plan

- Same multi-model test as the firm prompt
 - Run 10 times to check variability (the environment has more degrees of freedom)
 - Specifically validate:
 - units_sold sums correctly
 - market shares sum to 1.0
 - no production violations
 - no R&D advances when below threshold
 - Have a human read 5 narratives in sequence to check consistency
-

Iteration Strategy

Both prompts will need iteration. Common iterations:

If output JSON is invalid:

- Add explicit JSON schema
- Add an example response in the prompt
- Use OpenAI/Anthropic structured output if available
- Reduce prompt length (long prompts confuse smaller models)

If numbers are out of range:

- Add explicit "REASONABLE RANGE" hints in the prompt
- Show the deterministic baseline more prominently
- Add "common mistakes to avoid" section

If reasoning is incoherent:

- Add chain-of-thought structure ("First think about X. Then Y.")
- Ask for reasoning BEFORE the JSON output
- Reduce the number of decision dimensions

If output is too verbose:

- Add explicit "max 3 sentences" instruction
- Use temperature=0
- Truncate at LLM level

If output ignores context:

- Move critical context to the END of the prompt (recency bias)
 - Add explicit "you must reference X" instructions
 - Test with a stronger model first to confirm the prompt is right
-

After Manual Testing

When the prompts produce reliable output, the next step is to:

- Lock the prompt text in prompts.py as constants
- Build a `format_firm_prompt(firm_state, public_info, memory)` function that

fills in the dynamic sections

- Build the same for environment
- Add a `validate_response(json_str, schema)` function with retry logic
- Add unit tests with mock LLM responses

The prompts are then load-bearing. Changes to them should require re-running the test plan to verify nothing regresses.